VIDYAPEETH

BATCH CODE: 12-AN181MA

SUBJECT NAME: CHEMISTRY

CHAPTER NAME: Structure of atom

Lecture No. 10

By Amit Sir

Today's Goal

Subtopic .. Discovery of Neutron

Subtopic .. Thomson Model

Subtopic .. Rutherford atomic Model

Proton

* absolute mass of proton $= 1.672 \times 10^{-27}$ kg
* absolute charge on proton $= \boxed{+1.602 \times 10^{-19} C}$ $+e$
* specific charge on proton $= 9.57 \times 10^{7} \frac{C}{kg}$

Relative mass = 1
Relative charge = +1

Discovery of neutron: →

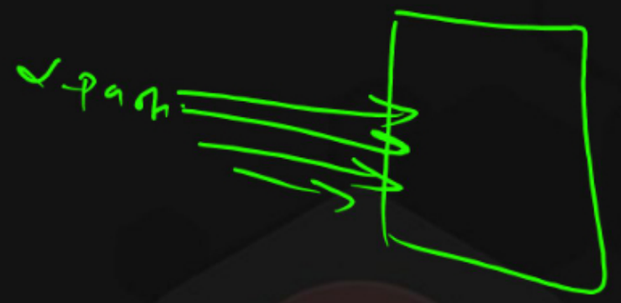
⇓

discovered by chadwick.

→ They bombarded α-particle on Berdium sheet.

He^{2+} → α-particle

$$^{4}_{2}He + ^{9}_{4}Be \longrightarrow ^{12}_{6}C + ^{1}_{0}n$$



absolute charge on neutron = 0

- absolute mass of neutron = 1.675×10^{-27} kg

- relative charge = 0
- relative mass = 1
- specific charge = $\frac{|\text{charge}|}{\text{mass}} = 0$

$1|e^{2+}$

$m_p \simeq m_n = 1837 m_e$

$m_p \simeq 1840 m_e$

sp. charge

electron → $\frac{e}{m_e}$

proton → $\frac{e}{m_p} = \frac{e}{1837 m_e}$

neutron → $\frac{0}{m_n} = 0$

α-Particle (He^{2+}) → $\frac{2e}{4m_p} = \frac{e}{2 \times 1837 m_e}$

sp. charge

electron > P > α-particle > neutron

Atomic Model

① Thomson atomic Model
or
Plum Pudding Model

→ atom is a uniform sphere of ⊕ve charge in which electron is embeded like seeds in Watermelon.

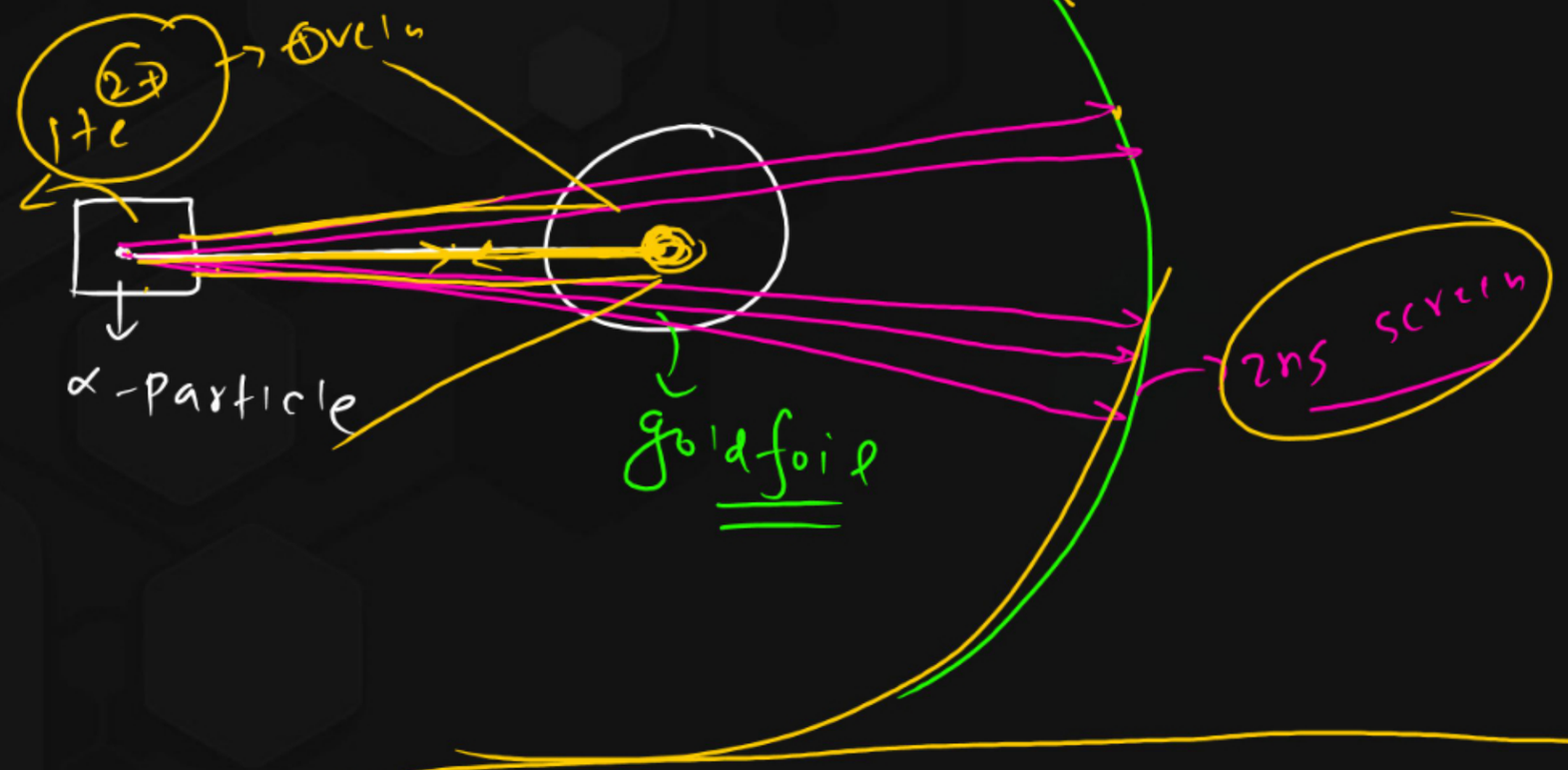
Drawback: ⇒

(i) electron is not static.

(ii) ⊕ve charge is not spreaded in whole atom.

(iii) It does not explain the stability of atom.

Rutherford atomic Model :→



observation

① since most of the α-particles passes undeflected it means most of the space of an atom is empty (Void)

② Some Particles are deflected by small angles it means +ve charge is present inside the atom.

③ one out of 20'000 Particle retraces their Path it means +ve charge is concentrated in Small volume and it is hard and rigid.

$$(1V_1)^+ = \frac{e}{23\,mp}$$

Topic: Homework-1

- DPP

DPP NO	Question No
DPP -	
DPP -	

- Module
  - Exercise Name

Module Page No	Question No
Page No-	
Page No -	

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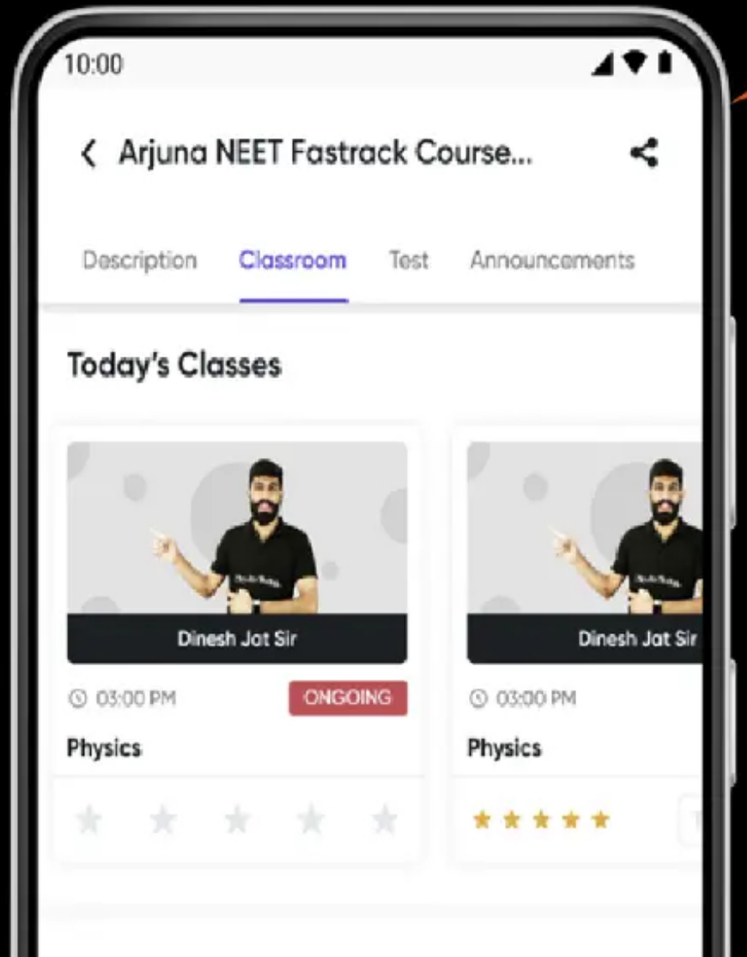
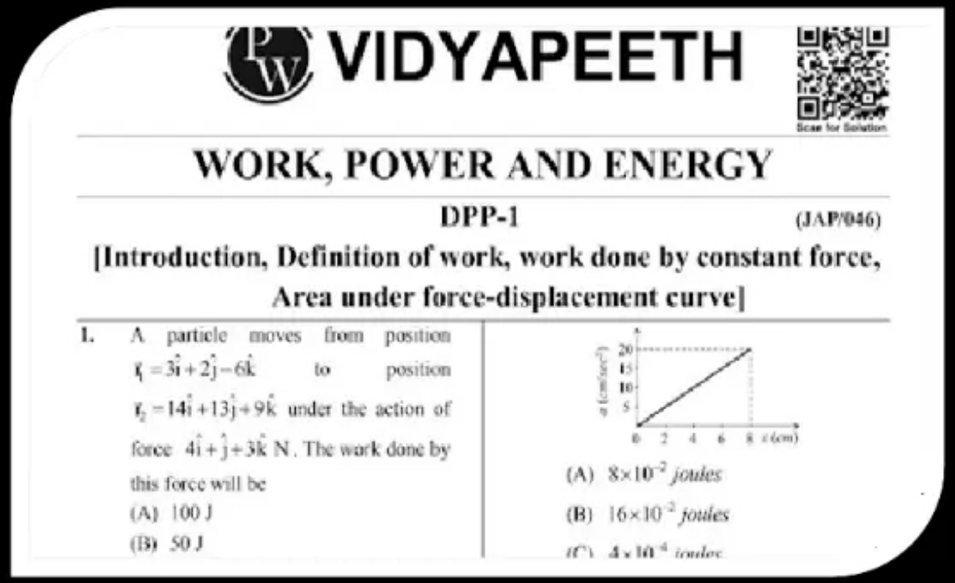


"SCAN" to join our "TELEGRAM" channel

Solve the DPP and check Solution



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Scan for Solution

WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} joules

(B) 16×10^{-2} joules

Thank You!!!!

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